

Andrew Yeung

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EDUCATION

University of Toronto

Sep 2024 – Apr 2029

BASc. in Electrical & Computer Engineering + PEY Co-op

Toronto, ON

- cGPA: **3.76/4.0**, Ranked **Top 30** in Electrical & Computer Engineering Department (2024 – 2025)
- Dean's Honour List (2024 Fall, 2025 Winter, 2025 Fall, 2026 Winter)
- Courses: Digital Systems (Verilog), Computer Organization (RISC-V Assembly), Introductory Electronics (LTspice)

EXPERIENCE

FPGA Research Intern

May 2026 – Present

Intelligent Sensory Microsystems Lab

Toronto, ON

- Developed **Verilog RTL** modules to control and interface a custom indirect Time-of-Flight CMOS image sensor for 3D depth sensing in Prof. Roman Genov's **ISML** group.
- Verified designs on the **XEM7310-A75 Artix-7™** FPGA using **Vivado** simulation tools, custom **Python** scripts, the **Integrated Logic Analyzer (ILA)**, and oscilloscopes to debug timing and signal integrity issues.
- Implemented a **MicroBlaze™** processor system in Vivado and **Vitis™** using **AXI4** interface protocol to enable communication between embedded software and FPGA peripherals.
- Recipient of a University of Toronto Engineering **Research Award** funding 4 months of summer research.

Da Vinci Engineering Enrichment Program Instructor

May 2025 – Aug 2025

University of Toronto Engineering Outreach Office

Toronto, ON

- Developed and instructed a hands-on program introducing hardware concepts to **Grades 3–8** students using **Arduino**, including breadboard prototyping, sensor integration (e.g., ultrasonic sensing), and motor control.
- Delivered engaging, inquiry-based instruction to diverse student groups, adapting content to different academic levels and learning styles to foster **critical thinking**, **creativity**, and **problem-solving**.

PROJECTS

Rhythm Racer | C, RISC-V, FPGAs (DE1-SoC) —

Mar 2026 – Apr 2026

- Developed an interactive speed rhythm typing game in **C** on a **RISC-V** soft-core processor mapped to the **DE1-SoC FPGA**, where users type song lyrics in sync with audio playback using **interrupt-driven I/O**.
- Implemented **VGA** graphics, **PS/2** keyboard interfacing, and audio output to display song lyrics, capture user input, and provide audiovisual gameplay.
- Integrated song data storage, real-time scoring, and live **WPM** tracking while leveraging **double buffering** and hardware **FIFOs** to ensure responsive gameplay and smooth audiovisual performance.

Sorting Algorithm Visualizer | Verilog, FPGAs, Quartus Prime, ModelSim —

Nov 2025 – Dec 2025

- Implemented a fully functional **bubble** and **insertion** sort simulator with graphics and user input using **Verilog**, the **DE1-SoC FPGA**, and a **VGA output monitor**.
- Used **finite state machines (FSMs)**, **registers**, and **sequential logic** to control algorithm flow, display dynamic sorting animations as bar graph rectangles, and handle real-time user interactions.
- Verified correctness through **ModelSim** and **DESIm** simulations and hardware testing, debugging **timing issues** and validating module logic before project compilation on **Quartus Prime**.

Wearable Speech Translator | Python, Arduino, APIs, TMRpcm —

Jan 2025 – Feb 2025

- Developed a real-time wearable speech translation device, integrating **microcontrollers**, **machine learning**, and **assistive technology design** to translate spoken input into multilingual audio output in real time.
- Implemented **Faster Whisper** for speech recognition and transcription, with **Arduino + TMRpcm** for audio I/O, **SD card storage**, **Google Translate API** for multilingual translation, and **Python serial communication** for hardware–software integration.

SKILLS

Languages: Verilog, C++, C, Java, Python, RISC-V Assembly, JavaScript, HTML/CSS

Developer Tools: Quartus Prime, ModelSim, Vivado, LTspice, Vitis, Git, Arduino IDE, DESIm, MATLAB

Professional Skills: Teamwork, Collaboration, Critical Thinking, Adaptability, Initiative, Leadership